

SCI-CAL

Detector Calibration, Atmospheric Extinction, and Signal Transfer

Engineering Conformance

Version 0.1 / Engineering revision r0.4 - DRAFT

2026-08-20

An implementation-independent conformance view of the same shared scientific authority, suitable for a clean implementation or future audit without mapping to current files, functions, classes, or behavior. This revision is the formal companion to Science-Team Rationale r0.5.

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Independent, implementation-blind authorship

Draft authority boundary. This is an implementation-blind scientific contract draft. It defines proposed scientific meaning and falsifiable conformance conditions. It is not evidence of implementation conformity, atmosphere-model fidelity, observational repeatability, absolute flux accuracy, production readiness, or completed validation.

Abstract

This document converts the shared SCI-CAL v0.1 scientific authority, as explained by Science-Team Rationale r0.5, into observable conformance conditions. A candidate realizes a typed, occurrence- scoped transformation from admitted **xs** samples to top-of-atmosphere point-source-peak mJy beam^{-1} , applies selected **flxscale** and target atmosphere correction exactly once, preserves the boundary to downstream measurement-noise estimation, maintains a complete systematic ledger, and emits resolvable lineage and truthful validity/quality state. This view does not prescribe present software structure and does not assert that any implementation conforms. The common numbered requirements are the only normative clauses; the procedures here describe evidence for them. Q01–Q09 are scientifically dispositioned; explicitly unavailable uncertainty products and final performance acceptance remain visible and cannot be fabricated by candidate metadata or implementation behavior.

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1 Conformance object and non-claims

The object under review is one complete SCI-CAL transformation for a coherent observation or declared segment, including its inputs, resolved state, realized numerical outputs, uncertainty products, validity/quality state, canonical lineage, and compact product links. A numeric array alone is not a conformance object.

Conformance is evaluated against the common definitions, equations, assumptions, requirements, and edge predictions included below. The review does not infer scientific meaning from names, defaults, row positions, file placement, or historical behavior. It does not map requirements to current files, functions, classes, configuration keys, or tests. Such a mapping, if later authorized, is a separate implementation audit.

This r0.4 engineering view is the semantic companion to Science-Team Rationale r0.5 and its Q01–Q09 decision register. Retaining every formal ID is necessary but not sufficient for consistency: scientific ownership, transfer validity, broadband convention, and ordering limitations must also survive into a conformance disposition.

A structural pass establishes only the first claim layer in Table 4. Atmosphere representation and observational performance require their own evidence. Its strongest structural-policy state is **science-qualification-eligible**, never an achieved **science-qualified** or **calibrated-science** claim. The latter claims remain governed by the separate-evidence threshold in SCI-CAL-REQ-049. No production state follows from this draft.

2 Logical conformance records

The following are logical records. They may be serialized or implemented in any form that preserves the stated semantics and relationships.

Table 1: Logical records to expose for conformance review.

Record	Minimum semantic contents	Primary checks
Requested state	requested channel, target unit/meaning, mode, source choices, and explicitly supplied expert values	immutable acceptance; unsupported requests remain visible
Effective state	resolved activation, legal unit path, normalized factor direction, selected operator policy, and rejection diagnostics	one-way resolution; “automatic” resolves to an exact effective choice
Observation-resolved state	observation/Tune, acquisition keys, column binding evidence, selected APT identity/digest, row occurrences, association edges, factors, atmosphere/time support, response basis, and uncertainty ledger	complete replacement per observation; unique keys and supported domains
Owner-decision snapshot	Q01–Q09 state, owning authority, required evidence, resolution source/date, decided semantics, explicitly unavailable products, and final owner-acceptance state	no local default, candidate assertion, or implementation behavior substitutes for approved authority

Record	Minimum semantic contents	Primary checks
Source-calibration record	Beammap/source-APT producer identity, source APT and row occurrence, calibrator model/epoch, source atmosphere, amplitude estimator, beam/template normalization, pointing treatment, spectral convention, factor direction, and uncertainty/covariance status	selected factor meaning and ancestry resolve to the producing authority
Selection/child-transform record	TolProj association, selected observation-specific child APT, parent source APT, approved transformation and version, transfer-validity domain, retained systematic, and exact child value	transformer changes only approved fields and never reinterprets producer quantities
Delivery record	TolTECA-delivered selected artifact identity/digest and immutable handoff identity	delivery changes neither scientific meaning nor factor value
Photometric-convention state	reference frequency or wavelength, reference spectrum, calibrator/color treatment, passband weighting/variation, and compatibility across source factor, target atmosphere, and reported unit	unsupported monochromatic or cross-array interpretation remains unavailable
Pipeline boundary/order state	ordinary dimensionless $\mathbf{x_s}$ boundary, CAL-before-PTC relation, and declared relation to detector mixing, PCA/common-mode removal, temporal filtering, weighting, mapmaking, and downstream response	CAL performs no additive operation; PTC owns DC and correlated-mode removal
Realized factor ledger	one entry per factor instance and occurrence/support, including role, value, unit, direction, reference plane, stage, parent lineage, validity, and application count	Equation (14); no embodied or parent double application
Realized output state	valid occurrence set, calibrated values, downstream conditional-uncertainty boundary, observation quality, reason codes, and response status	Equations (13), (17), and (1)
Canonical lineage	all resolvable identities and relationships in Equation (28) plus nuisance completeness/correlation scope	deterministic reconstruction and no stale observation state
Product link	canonical-record reference, selected APT reference, and occurrence keys sufficient to resolve both	resolvable without copying the entire per-detector table

3 Conformance evaluation sequence

The sequence below is explanatory rather than an implementation architecture. It groups the observables that the common requirements place at the cold boundary before a hot sample calculation can have scientific meaning.

- E1. Resolve authority gates.** Load the exact Q01–Q09 snapshot and apply its decided semantics and each explicitly unavailable product to the requested output or claim. A candidate cannot promote missing numerical uncertainty or unaccepted performance to an approved claim.
- E2. Normalize the request.** Establish that the measured stream is `xs`, the target is top-of-atmosphere point-source-peak mJy beam^{-1} , the physical input and photometric-convention states are declared, and the requested/effective distinction is recorded.
- E3. Resolve occurrence identity.** Validate acquisition keys, the column relation, source and selected child APT identities, local row occurrences, target-to-source edges, approved transformation, delivery identity, and array/network constraints. Establish exact cardinality for the selected support before evaluating factors.
- E4. Resolve factor causality.** Admit one selected `flxscale` per detector occurrence, resolve its producer-owned generating record, verify its direction and reference plane, classify the child transformation and pointing-correction ancestry, and reject opaque or duplicate factor roles.
- E5. Resolve atmosphere support.** Validate opacity and airmass meaning, WVR time coverage/bracketing, whole-observation quality, exact content-bound operator and digests, node support, log-transmission orientation, elevation and opacity interpolation, and passband identity/limitations.
- E6. Resolve ordering and broadband compatibility.** Bind the declared no-additive-operation CAL boundary and CAL-before-PTC order. Verify the 272/214/150 GHz per-array convention, source spectrum, exact array-average atmosphere passband, and downstream ownership of target color correction.
- E7. Evaluate once.** Only on support admitted by the applicable authority gates, evaluate Equations (7), (12), and (13). Record factor-instance application counts and numeric factor ranges. Invalid occurrences remain outside the domain rather than becoming zeros.
- E8. Preserve the uncertainty boundary.** CAL creates no statistical variance or weight. Verify that any downstream companion on the calibrated support obeys Equation (17) or (18), and that unavailable producer/systematic terms remain unavailable rather than zero.
- E9. Resolve nuisance and response claims.** Complete the required nuisance ledger, retain correlations, link the originating Beammap response separately from any realized downstream response, and withhold total or point-source-peak claims that lack their conditions.
- E10. Publish reconstructible state.** Emit the validity tuple, reason codes, one canonical lineage record, and resolvable product links. State the status of structural, representation, and observational claims separately.

4 Scientific-owner dispositions and remaining gates

Table 2 carries the Science-Team Rationale r0.5 decision register into engineering disposition. Q01–Q09 are decided; a decision may still require an explicit unavailable numerical product or a future owner acceptance based on evidence.

Table 2: Owner decisions and required engineering consequences.

ID	Approved disposition	Required engineering consequence
SCI-CAL-OWNER-Q01	Ordinary <code>xs</code> is dimensionless $\Delta f/f_{\text{res}}$, positive with optical power, with no CAL baseline	Validate Tune and expose the half-resonance-width high-noise condition.
SCI-CAL-OWNER-Q02	CAL precedes PTC; PTC owns DC/common-mode removal	Preserve this order and perform no additive CAL operation.
SCI-CAL-OWNER-Q03	Frozen SCI-BEAM derivation of <code>flxscale</code>	Resolve the complete producer record; retain source-factor uncertainty as unavailable until supplied.

ID	Approved disposition	Required engineering consequence
SCI-CAL-OWNER-Q04	Closest accepted Beammap APT by default; optional scientist-directed TolProj per-array rescale	Materialize an immutable child APT and record its reference evidence; retain rescale uncertainty as unavailable until defined.
SCI-CAL-OWNER-Q05	272/214/150 GHz convention; source-dependent spectrum; downstream target color correction	Bind the declared spectrum/passband and expose unavailable detector/network passband variation.
SCI-CAL-OWNER-Q06	Exact WVR/AM/passband operator and content digests	Evaluate within $0 \leq \tau_{225} \leq 0.25$, $25^\circ \leq EL \leq 80^\circ$, and the authorized α set; fail closed outside.
SCI-CAL-OWNER-Q07	One CAL class per observation, 0.025 momentary-excursion tolerance	Never split an observation; separate quality classification from sample-level numeric support.
SCI-CAL-OWNER-Q08	Noise downstream; CAL ultimately owns array-dependent systematics	Publish current missing source, rescale, WVR, and covariance terms as unavailable; do not emit total uncertainty/significance.
SCI-CAL-OWNER-Q09	Concrete Beammap closure and associated-pointing transfer workflow	Report achieved evidence by array without arbitrary sample-size/matrix gates; final scientific acceptance remains an owner decision.

5 Notation and typed quantities

The contract uses occurrence-scoped identities. An index is never, by itself, a persistent physical identity. Let o denote an observation or Tune context, t a location on its approved common sample axis, d a selected detector occurrence, $a(d)$ its TolTEC array, and $j = (o, t, d)$ a compound sample occurrence. The three arrays are $a1100$, $a1400$, and $a2000$. Networks 0 through 6 belong to $a1100$, 7 through 10 to $a1400$, and 11 through 12 to $a2000$.

Table 3: Canonical symbols and types.

Symbol	Identity and support	Unit	Meaning
x_j^{xs}	acquisition sample occurrence	dimensionless	ordinary fitted KID resonance-frequency change $\Delta f / f_{\text{res}}$
$z_j = x_j^{xs}$	valid sample support	dimensionless	CAL input; no additive CAL baseline operation
K_{od}^{acq}	target acquisition occurrence	identity only	observation/Tune, network or interface, and network-local tone or row slot
R_{sd}^{src}	producer-owned source-APT row occurrence	identity only	immutable source APT identity plus source observation/network context and artifact-local row key

Symbol	Identity and support	Unit	Meaning
$E_{od,sd}^{\text{assoc}}$	target-to-source occurrence edge	identity only	matcher method/version, disposition, and available quality evidence
R_{od}^{child}	selected child-APT row occurrence	identity only	immutable observation-specific child artifact, parent relation, and artifact-local row key
G_{sd}^{src}	source-calibration generating record	identity and typed fields	producer, calibrator, source atmosphere, estimator, response, spectral convention, direction, and covariance lineage
$T_{od,sd}^{\text{child}}$	approved child transformation	identity plus typed rule	TolProj association/transformation, validity domain, retained systematic, and identity-transform state
D_o^{del}	selected-artifact delivery record	identity only	TolTECA handoff identity that preserves child meaning and value
F_{od}^{sel}	selected child-APT row occurrence	$\text{mJy beam}^{-1} (\Delta f / f)^{-1}$	authoritative absolute flxscale for this selected lineage
P_{od}	lineage event, if applicable	dimensionless	TolProj pointing-derived correction already embodied in F^{sel} ; never an additional CAL runtime factor
r_{od}	selected APT row occurrence	declared relative-response unit	responsivity ; a relative quantity, not absolute flux calibration
s_{od}	selected APT row occurrence	declared sensitivity unit	Beammap sens ; sensitivity or approximate-weight input only
$\tau_{225,o}(t)$	declared time support	dimensionless	zenith optical depth at 225 GHz
X_{ot}	eligible sample occurrence	dimensionless	full sample airmass supplied under approved upstream meaning
$\ell_{ot} = \tau_{225,o}(t)X_{ot}$	eligible sample occurrence	dimensionless	line-of-sight 225 GHz optical-depth coordinate relative to $X_{\text{ref}} = 0$
$\mathcal{O}_a$	content-bound operator for array a	identity plus nodes	retained am12_fixed_djf25_pieewise_linear_los_tau_v1 operator
$\lambda_{a,\alpha}(\tau, EL)$	operator support	dimensionless	authoritative $-\ln T^{\text{eff}}$ surface
$T_{a,\alpha}(\tau, EL)$	operator support	dimensionless	broadband array transmission
$C_{a,\alpha} = T_{a,\alpha}^{-1}$	operator support	dimensionless	positive correction from the eligible sample plane to top of atmosphere
M_j	valid sample occurrence	$\text{mJy beam}^{-1} / U_{x,d}$	exact realized once-only multiplier
y_j	valid sample occurrence	mJy beam^{-1}	calibrated top-of-atmosphere, point-source-peak detector sample

Symbol	Identity and support	Unit	Meaning
$C_{x \theta}$	valid input vector support	U_x^2 with typed blocks	conditional measurement covariance at fixed calibration state θ
$C_{y \theta}$	identical valid output support	$(\text{mJy beam}^{-1})^2$	transformed conditional measurement covariance
$v_j^{(x)}, w_j^{(x)}$	scalar sample occurrence	U_x^2, U_x^{-2}	diagonal conditional variance and inverse variance, when supplied
θ	named detector, array, observation, cohort, or global scope	parameter-specific	calibration, atmosphere, and response nuisance state
Π_o	observation/segment	identity plus typed fields	photometric-convention compatibility and unavailable status
Ω_o	observation/segment	identity plus typed fields	pre-CAL boundary and declared global pipeline-order state
Q_o^{owner}	package/observation authority snapshot	identity plus state	Q01–Q09 ownership, evidence, resolution, and blocked-claim state

Bold symbols stack only the declared valid occurrences; they do not insert numeric zeros for invalid samples. A locator such as a global column, table row, or map index may find an occurrence, but cannot establish its identity.

6 Authoritative definitions and claim boundaries

6.1 Operation and endpoint

SCI-CAL v0.1 operation. On admitted **xs** support, SCI-CAL binds each target acquisition occurrence to one row occurrence in one exact selected immutable observation-specific child APT, applies that row’s absolute **flxscale** once without reinterpreting its producer-owned meaning, applies one supported target-observation atmosphere correction to the top-of-atmosphere plane, and records the present availability of calibration systematics. Statistical measurement-noise estimation remains downstream. It ends at calibrated detector samples, not at a map, fitted source, or integrated photometric estimate.

Point-source-peak mJy beam^{-1} . The label denotes a peak-normalized response convention. For an unresolved source of top-of-atmosphere flux density S mJy, a downstream map whose realized beam, mapmaker, kernel, and filter response has unit peak for the declared response basis has ideal peak S mJy beam^{-1} . This definition does not authorize surface-brightness, per-pixel, temperature, extended-source, or integrated-flux interpretation.

Reference plane. The target is top of atmosphere with $X_{\text{ref}} = 0$. A target sample at airmass X_{ot} is corrected from its observing plane to that reference. Atmosphere corrections used while deriving the selected source APT belong to that APT’s ancestry; they do not replace or duplicate the target-observation correction.

6.2 Identity and immutable selection

Acquisition occurrence. The tuple of observation/Tune context, network or interface, and network-local channel/tone slot. It identifies the target measured occurrence to which a sample column is bound.

Measured-APT row occurrence. The tuple of immutable selected-artifact identity, its artifact-local row key, and source observation/network context. A UID is local to that immutable artifact. It is not a universal detector name.

Source APT and observation-specific child APT. The source APT is the producer-owned Beammap calibration artifact whose row supplies the scientific meaning and generating lineage of `flxscale`. The selected child APT is the exact immutable observation-specific artifact chosen or produced by TolProj from an admitted target-to-source association and, if applicable, one explicitly approved transformation. An identity transformation is recorded when the source row is selected unchanged. Parent and child occurrences remain distinct even when their values are numerically equal.

Association edge. A separate, directed target-to-source relation that names both occurrence-scoped endpoints, matcher method and version, disposition, and available quality evidence. Only a unique admitted match is sufficient for calibration. Aggregate matcher performance is not a per-row match probability.

Identity layers. Acquisition identity, measured-APT identity, cross-observation association, and optional design identity answer different questions. Occurrence identity, order-independent semantic-content identity, and exact byte-transport identity are also distinct. Equal row numbers, local keys, paths, timestamps, or integer spellings do not prove correspondence.

Binding. Either an explicit keyed relation or a verified ordered-row contract may bind acquisition occurrences to sample columns and selected APT rows. An ordered relation is valid only with recorded scope, cardinality, ordering definition, and proof. Row position alone is not identity.

Producer-transformer-delivery-consumer boundary. Beammap/source-APT production owns the source calibration, source atmosphere, estimator, beam/template normalization, pointing treatment, spectral convention, uncertainty, and published factor meaning. TolProj owns target-to-source association and only an explicitly approved child transformation with its lineage. TolTECA may deliver the selected immutable artifact into Citlali but does not change its scientific meaning or value. SCI-CAL consumes the exact selected child value, applies it once with target atmosphere, and does not rederive or reinterpret producer or transformer quantities.

6.3 Factors and once-only lineage

Factor instance. A value together with a stable instance identity, scientific role, direction, unit, recipient identity, support, reference plane, validity, application stage, uncertainty status, and parent lineage. Numerically equal values with different roles are not interchangeable.

Selected absolute factor. F_{od}^{sel} is the selected child-APT row's `flxscale`, oriented under the contract as $\text{mJy beam}^{-1}/U_{x,d}$. It is the sole absolute detector factor in the v0.1 signal multiplier. Its scientific meaning resolves to the source-calibration producer record. If TolProj created a transformed child APT, every approved transformation, including an embodied pointing-derived correction, is an ancestry event already represented in this value. The source-factor meaning is supplied by SCI-BEAM, the transfer policy and any optional per-array pointing-source photometric rescale are supplied by TolProj, and the selected child artifact records their lineage.

Relative responsivity. `responsivity` may support a separately declared relative-response, diagnostic, normalization, or nuisance role. It does not independently establish an absolute mJy beam^{-1} scale and is not an additional v0.1 signal multiplier.

Sensitivity. Beammap `sens` is a separately typed sensitivity or approximate-weight quantity. It is neither atmospheric extinction nor an absolute signal factor nor a general total uncertainty.

Opaque total factor. An unqualified `fcf` is not authoritative. A compatibility value can be admitted only when its exact contents, units, direction, exclusions, and constituent instance identities reproduce the canonical factorization without omission or duplication.

Once-only composition. Each required physical role contributes once to a realized multiplier. Causal derivation order is recorded in lineage; numeric multiplication of admitted scalar factors is order independent. A factor already materialized into a selected artifact is not multiplied again at execution.

6.4 Atmosphere operator and passband status

Atmosphere operator. The content-bound operator is

`am12_fixed_djf25_pieewise_linear_los_tau_v1.`

It uses the fixed `LMT_DJF_25.amc` model profile, an analytic zero-opacity anchor, shape-preserving PCHIP in elevation at each opacity anchor, and piecewise linear interpolation of effective line-of-sight optical depth across opacity anchors. It returns $T = \exp(-\lambda)$ and $C = T^{-1}$, has no operator switch at $\tau_{225} = 0.15$, and is supported on $0 \leq \tau_{225} \leq 0.25$ and $25^\circ \leq EL \leq 80^\circ$. Its node-table digest is

```
fd688a4cd3f46585b08631bc63a562aed482feb9b24ec9ee0071b70db7eb8a5f.
```

Passband reference. The only selected passband identity is

```
toltec-passband-set-v1:sha256:
5e6f38f14bcae93a29ffe8362c52b15209f51aee4e48373b23aaa5ec2f8a6433.
```

It binds one array-named modeled curve for each TolTEC array and an index. The frequency coordinate is in GHz and the supplied throughput is dimensionless. The declared operator uses the supplied throughput as an energy-weighted array-average response and normalizes its spectral integral by the same response/source-spectrum integral. Detector/network variation, telescope measurement, and passband uncertainty/covariance are unavailable. The passband identity and numerical representation do not establish atmospheric or telescope-response truth.

Photometric-convention state. The per-array reference frequencies are 272 GHz (a1100), 214 GHz (a1400), and 150 GHz (a2000). A declared reference spectrum $S_\nu \propto \nu^\alpha$ selects one precomputed atmosphere surface for $\alpha \in \{-1, 0, 2, 4\}$, defaulting to zero; alpha interpolation and extrapolation are prohibited. BEAM supplies one per-array mJy beam⁻¹ source value to `flxscale`. SCI-CAL applies no second calibrator color correction. Target-source color correction is a downstream scientist action.

Pipeline boundary/order state. SCI-CAL acts multiplicatively on the ordinary `xs` stream with no additive baseline operation. PTC follows CAL and owns DC removal and correlated-atmosphere/common-mode cleaning. MAP/FLT still owns the realized downstream response.

6.5 Uncertainty and response

Conditional measurement uncertainty. Statistical measurement-noise variance, weight, or covariance is estimated downstream, not by SCI-CAL. CAL does not synthesize or publish such a product. The fixed-multiplier covariance identity remains a downstream consistency relation, not a present CAL output.

Nuisance uncertainty. Uncertainty in calibration, atmosphere, or response state, with named correlation scope. It is not included in a conditional weight unless explicitly incorporated through a documented joint covariance. Missing nuisance uncertainty is unavailable, not zero.

Response basis. The originating Beammap APT beam/template occurrence, including elliptical parameters and their units/frame when available, together with the separately supplied realized mapmaker, kernel, and filter response required to interpret a downstream point-source peak. A circularization is a labeled approximation, never silent replacement.

6.6 Validity and quality tuple

An immutable owner-decision snapshot accompanies the tuple below. It records the decided Q01–Q09 policies, their resolution source/date, and every still-unavailable numerical product or achieved claim.

Every observation has a state tuple

$$\mathcal{V} = (R, E, S, Q, U_c, U_n, B), \quad (1)$$

where R is requested mode, E effective mode, S calibrated-signal validity, Q one observation-level opacity quality class, U_c conditional uncertainty availability, U_n nuisance/total-uncertainty completeness, and B response-basis completeness. Required reason codes accompany every non-success state. In particular:

- **science-qualification-eligible** means structurally admissible under the v0.1 policy with fully resolved scientific inputs. It is an eligibility state, not an achieved **science-qualified** or **calibrated-science** claim. Unless the owner later defines a different explicit threshold, either achieved claim additionally requires separately accepted atmosphere-representation-fidelity and observational-performance evidence over the same declared identity and support;

- **engineering-availability** means the observation is retained for engineering use under the same supported correction operator, with no science-quality calibration claim;
- **outside-supported calibration** means the request lies outside an admitted domain or lacks a required operator/input identity;
- **disabled/uncalibrated** records an intentional effective no-op and never relabels uncalibrated numbers; and
- **uncertainty unavailable** may coexist with a valid calibrated signal when the missing item is not required to define the signal, but it prohibits the corresponding uncertainty or significance claim.

6.7 Three claim layers

Table 4: Claims that must not be conflated.

Claim	What it establishes	What it does not establish
Structural calibration correctness	applicable owner gates, typed inputs, producer/transformer/delivery/consumer lineage, valid association, exact factor algebra, once-only lineage, supported operator evaluation, downstream uncertainty boundary, and reconstructible output	truth of the atmosphere model, universal transfer, telescope response, repeatability, absolute flux accuracy, or production readiness
Atmosphere representation fidelity	agreement of the declared finite operator with a named higher-fidelity atmosphere calculation over declared support	agreement with the actual atmosphere, WVR accuracy, telescope passband truth, or source recovery
Observational calibration performance	repeatability and absolute recovery against independent astronomical evidence with full covariance	structural correctness by itself or universal performance outside the studied population and support

The prior one-percent representation, five-percent repeatability, and five-to-ten-percent absolute-accuracy figures are reporting benchmarks, not automatic pass/fail ceilings or guarantees. Achieved behavior is reported honestly for the available closure and transfer observations.

7 Assumptions, conditions, and unavailable authority

The equations are conditional on the following explicit statements. They are not permissions to replace missing information with defaults.

SCI-CAL-ASM-001: upstream sample authority.

The supplied sample axis, time, eligibility, original/synthesized state, elevation/airmass meaning, and acquisition-column binding are correct under approved upstream authority. Ordinary $\mathbf{xs}$ is dimensionless KID $\Delta f/f_{\text{res}}$, positive with increasing absorbed optical power, with no absolute DC sensitivity. The fitted resonance is expected to match the active probe tone. Wrong-Tune and high-noise quality states are explicit.

SCI-CAL-ASM-002: ordinary measured channel.

The admitted signal is the ordinary $\mathbf{xs}$ stream. A distinct measured stream is a distinct physical observable and does not inherit this factor.

SCI-CAL-ASM-003: selected immutable child APT.

Beammap/source-APT production owns the generating calibration record. TolProj has supplied one admitted target-to-source association and selected or created one immutable observation-specific child APT, with an identity transformation recorded when the source row is unchanged. TolTECA delivery preserves that exact artifact. SCI-CAL does not choose the artifact, matcher, or transformation and does not reinterpret the selected value.

SCI-CAL-ASM-004: no additive CAL operation.

SCI-CAL performs no baseline or offset subtraction. PTC is downstream and owns DC removal and correlated-atmosphere/common-mode cleaning.

SCI-CAL-ASM-005: coefficient direction and reference plane.

The selected `flxscale` is finite, nonzero, oriented as $\text{mJy beam}^{-1}/(\Delta f/f)$, and refers to the top-of-atmosphere calibration plane. Its producer record states calibrator model/epoch, source atmosphere, amplitude estimator, beam/template normalization, pointing and spectral treatment, direction, and uncertainty/covariance status. Its child lineage states the approved transformation, transfer-validity domain, retained systematics, and disposition of any pointing-source photometric rescale. SCI-BEAM owns the generating source-factor meaning; TolProj owns only the admitted child transformation and its lineage.

SCI-CAL-ASM-006: atmosphere inputs.

τ_{225} is dimensionless zenith optical depth from the LMT WVR values written into the observation's `tel*.nc` file. The approximately five-minute readings are interpolated onto sample time. X is the full eligible-sample airmass and the target pivot is $X_{\text{ref}} = 0$.

SCI-CAL-ASM-007: fixed conditional state.

Conditional covariance equations hold at fixed identities, factor values, atmosphere state, selection, and response state. Dependence of fitted state on the same signal requires cross-covariance or joint propagation.

SCI-CAL-ASM-008: response ownership.

Beammap/source-APT production supplies the originating beam/template and source-calibration identity; MAP/FLT supplies the realized mapmaker/kernel/filter response. SCI-CAL records and conditions on those identities but does not certify their empirical fidelity or reinterpret their producer-owned meaning.

SCI-CAL-ASM-009: coherent observation class.

CAL assigns one opacity quality class to the complete observation and does not split an observation into science and engineering pieces. A 0.025 tolerance prevents momentary excursions from invalidating the observation; it does not extend the numerical operator's sample support.

SCI-CAL-ASM-010: passband limitation.

The selected TolTECA v1 passband set is a modeled array-named reference with the recorded unknowns in Section 6. The operator uses its supplied energy-weighted throughput with denominator normalization and the selected reference spectrum. Passband uncertainty and detector/network variation remain unavailable.

SCI-CAL-ASM-011: content-bound numeric operator authority.

The adopted machine contract and node table for `am12_fixed_djf25_piecewise_linear_los_tau_v1` are content-bound by:

```
7a064ff768a3de4f427f1338d94ef6cb9026d248f3c3c816fc3dfc96d156e36a
fd688a4cd3f46585b08631bc63a562aed482feb9b24ec9ee0071b70db7eb8a5f.
```

The first is the contract digest and the second the node-table digest. This authorizes numerical representation on declared support; it does not establish WVR truth, observational accuracy, or scientific acceptance.

8 Canonical equations and invariants

8.1 Admission and valid support

Let A_j be the conjunction of the scientific admission predicates: the sample is upstream-eligible `xs`; its acquisition key and sample column binding are resolved; its source and selected-child APT rows, unique admitted association, approved transformation, and delivery identity are resolved; applicable owner gates admit the requested physical output; units, factor directions, photometric convention, and pipeline boundary are declared; the sample, Tune state, factor, opacity, airmass, time support, operator, and response basis are valid for the requested operation; and the effective mode is enabled. The mathematical domain is

$$\mathcal{J}_{\text{valid}} = \{j : A_j = \text{true}\}. \quad (2)$$

Restriction to $\mathcal{J}_{\text{valid}}$ is not multiplication by a Boolean mask. Invalid occurrences do not become scientific zeros and do not contribute hits or coverage.

The owner-declared CAL input has no additive operation:

$$z_j = x_j^{xs}, \quad j \in \mathcal{J}_{\text{valid}}. \quad (3)$$

PTC acts downstream and owns removal of DC and correlated atmospheric/common modes.

8.2 Content-bound atmosphere operator

Let $\hat{\tau}_{225,o}(t)$ be the time interpolation of valid bracketing LMT WVR readings recorded in `tel*.nc`. The physical line-of-sight coordinate at the top-of-atmosphere pivot is

$$\ell_{ot} = \hat{\tau}_{225,o}(t) [X_{ot} - X_{\text{ref}}] = \hat{\tau}_{225,o}(t) X_{ot}, \quad X_{\text{ref}} = 0. \quad (4)$$

This is a coordinate for the retained operator, not a claim that a TolTEC band opacity equals τ_{225} , nor permission to replace the retained operator by an unapproved monochromatic exponential.

For array response $R_a(\nu)$, reference spectrum $S_\alpha(\nu) \propto \nu^\alpha$, and AM transmission $T_{\text{AM}}(\nu; \tau, EL)$, the frozen generating convention is

$$T_{a,\alpha}^{\text{eff}}(\tau, EL) = \frac{\int R_a(\nu) S_\alpha(\nu) T_{\text{AM}}(\nu; \tau, EL) d\nu}{\int R_a(\nu) S_\alpha(\nu) d\nu}. \quad (5)$$

The exact TolTECA-v1 ECSV nodes and supplied throughput are integrated by composite trapezoid with no spectral extrapolation. Define the authoritative ordinate $\lambda_{a,\alpha} = -\ln T_{a,\alpha}^{\text{eff}}$. At each opacity anchor it is evaluated by shape-preserving PCHIP in elevation. Between ordered opacity anchors τ_k and τ_{k+1} ,

$$u = \frac{\tau - \tau_k}{\tau_{k+1} - \tau_k}, \quad (6)$$

$$\lambda_{a,\alpha}(\tau, EL) = (1 - u)\lambda_{a,\alpha}(\tau_k, EL) + u\lambda_{a,\alpha}(\tau_{k+1}, EL). \quad (7)$$

The canonical transmission and correction are

$$T_{a,\alpha} = \exp[-\lambda_{a,\alpha}], \quad C_{a,\alpha} = \exp[\lambda_{a,\alpha}] = T_{a,\alpha}^{-1}. \quad (8)$$

The structural invariants are

$$T_{a,\alpha}(0) = C_{a,\alpha}(0) = 1, \quad 0 < T_{a,\alpha} < 1, \quad 1 < C_{a,\alpha} < \infty \quad \text{for positive supported opacity}, \quad (9)$$

with T_a strictly decreasing and C_a strictly increasing on declared support. Thus there is no positive-opacity unity plateau. Exact node values are reproduced and the operator is continuous at every seam. Its derivative may change discontinuously at a seam.

The authorized structural domain for a sample is the intersection

$$\mathcal{D}_{otd} = \{0 \leq \hat{\tau}_{225,o}(t) \leq 0.25, \quad 25^\circ \leq EL_{ot} \leq 80^\circ, \quad t \in \mathcal{D}_\tau, \quad \alpha \in \{-1, 0, 2, 4\}\}, \quad (10)$$

where $\mathcal{D}_\tau$ is the WVR source's declared time support. A time-resolved opacity at a sample is admissible only when measured at that time or obtained by a source-authorized interpolation between bracketing states whose declared validity covers the sample. No extrapolation from an unbracketed endpoint is allowed. Equation (10) is exact as a support intersection and is numerically instantiated only by the content-bound record in `SCI-CAL-ASM-011`. Observation-level quality classification and its 0.025 excursion tolerance do not expand this sample-level numerical support.

8.3 Once-only signal factorization

The selected observation-specific child APT is the value boundary for absolute calibration. Its value and scientific meaning resolve through the source-calibration generating record, admitted association, approved child transformation, and immutable delivery record. If TolProj has materialized a pointing-derived correction, its causal ancestry may be written

$$F_{od}^{\text{sel}} = P_{od} F_{od}^{\text{parent}} \quad (\text{lineage identity only}). \quad (11)$$

The right-hand factors are not separately applied to the target signal. Admissible correction dispositions are `not_applicable` and `embodied_once`, with the parent and event identities present for the latter. `required_but_unresolved` is not an admissible calibrated state.

Because v0.1 has only the target unit mJy beam^{-1} , its unit-transfer factor is the explicit identity $Q_{\text{mJy beam}^{-1}} = 1$. The exact realized multiplier and signal are

$$M_j = F_{od}^{\text{sel}} C_{a(d),\alpha}(\hat{\tau}_{225,o}(t), EL_{ot}) Q_{\text{mJy beam}^{-1}}, \quad (12)$$

$$y_j = M_j z_j, \quad j \in \mathcal{J}_{\text{valid}}. \quad (13)$$

No independent **responsivity**, **sens**, parent **flxscale**, pointing-derived correction, or opaque **fcf** appears in Equation (12). Any such quantity used upstream may be named in lineage or a nuisance model, but does not become another signal factor.

Numeric multiplication in Equation (12) is commutative. Causal provenance is not: the source APT and generating record precede the admitted association and child transformation; the exact selected child APT must exist before its row factor is delivered and bound; target atmosphere state is then applied at the target sample. For each factor role $\rho \in \{\text{absolute flxscale, target atmosphere, unit identity}\}$, define the realized application count $n_{j\rho}$. The once-only invariant is

$$n_{j\rho} = 1 \quad \text{for every required role and } j \in \mathcal{J}_{\text{valid}}, \quad (14)$$

and zero for roles not in the canonical multiplier. The factor-instance identities, not merely their numeric values, establish this count.

For a nonzero intended factor f , challenge results relative to the correct output obey

$$\frac{y_{\text{omitted}}}{y_{\text{correct}}} = f^{-1}, \quad \frac{y_{\text{duplicated}}}{y_{\text{correct}}} = f, \quad \frac{y_{\text{inverted}}}{y_{\text{correct}}} = f^{-2}, \quad (15)$$

when all other quantities are fixed and the inverse replaces the intended factor. A unity factor leaves values unchanged but must still retain its role and instance identity when the role is required.

8.4 Conditional measurement uncertainty

Stack the valid samples and let $A = \text{diag}(M_j)$. Conditioned on fixed state θ ,

$$\mathbb{E}[\mathbf{y} \mid \theta] = A\mathbb{E}[\mathbf{x}^{xs} \mid \theta], \quad (16)$$

$$C_{y|\theta} = AC_{x|\theta}A^\top. \quad (17)$$

This transformation preserves every supplied sample, detector, and observation correlation. For a diagonal conditional representation,

$$v_j^{(y)} = M_j^2 v_j^{(x)}, \quad w_j^{(y)} = \frac{w_j^{(x)}}{M_j^2}. \quad (18)$$

The valid support is identical to the signal support. If the corresponding input uncertainty is absent, the output state is unavailable rather than a zero variance or infinite weight.

Every synthetic injection, conditional noise realization, or genuine raw-domain Jacobian companion $k_{jr} = \partial x_j / \partial \alpha_r$ that is declared to share the signal response obeys

$$k_{jr}^{(y)} = M_j k_{jr}^{(x)}. \quad (19)$$

This equation does not authorize relabeling a physically distinct measured stream as a response companion.

8.5 Calibration and response nuisance covariance

The following equations are downstream consistency identities for a future systematic-uncertainty mechanism; SCI-CAL r0.5 does not yet publish their numerical inputs or a total covariance. For a fixed factor sign, let

$$H_{jr} = \frac{\partial \ln |M_j|}{\partial \theta_r}. \quad (20)$$

Within one differentiable patch of the retained two-coordinate atmosphere surface, define the local log slopes

$$\gamma_{a,\alpha}^\tau = \frac{\partial \ln C_{a,\alpha}}{\partial \tau_{225}}, \quad \gamma_{a,\alpha}^{EL} = \frac{\partial \ln C_{a,\alpha}}{\partial EL}. \quad (21)$$

For mean calibrated signal $\boldsymbol{\mu}$, the first-order nuisance term is

$$C_{\text{nuis}} \simeq \text{diag}(\boldsymbol{\mu}) H C_{\theta} H^{\top} \text{diag}(\boldsymbol{\mu}). \quad (22)$$

It is generally dense. A common fractional absolute-scale uncertainty s gives

$$C_{\text{common}} = s^2 \boldsymbol{\mu} \boldsymbol{\mu}^{\top}, \quad (23)$$

a rank-one contribution that does not decrease merely because more affected samples are averaged.

If fitted calibration state depends on the same data, a first-order total covariance also contains cross terms:

$$C_y \simeq A C_x A^{\top} + J_{\theta} C_{\theta} J_{\theta}^{\top} + A C_{x\theta} J_{\theta}^{\top} + J_{\theta} C_{\theta x} A^{\top} + \text{other declared cross terms}, \quad (24)$$

Independence can remove a cross term only when justified. Linearization is unavailable at an operator seam, for uncertainty spanning multiple intervals, for large or asymmetric multiplicative uncertainty, or for discrete association and selection uncertainty. Those cases require nonlinear samples, a joint posterior, or a bounded sensitivity study.

8.6 Response condition for the unit label

Let $P_B(\boldsymbol{\omega})$ be the originating Beammap response template, with declared coordinates and unit peak,

$$P_B(\mathbf{0}) = 1. \quad (25)$$

For an unresolved source of top-of-atmosphere flux density S mJy, a downstream realized linear response L has ideal map peak

$$m_{\text{peak}} = S \rho_L \text{ mJy beam}^{-1}, \quad \rho_L = [L P_B](\mathbf{0}). \quad (26)$$

The original point-source-peak label remains scientifically literal only when $\rho_L = 1$ on the declared support. If a kernel or filter changes ρ_L , the consumer must either propagate an explicit response renormalization and update the response-basis provenance, or withhold the literal peak-calibration claim. SCI-CAL does not estimate ρ_L .

For an explicitly selected elliptical Gaussian approximation, the originating solid angle would be

$$\Omega_B = \frac{\pi \theta_{\text{maj}} \theta_{\text{min}}}{4 \ln 2}, \quad (27)$$

with angle units, frame, position angle, and approximation label retained. Equation (27) records response basis only. It does not authorize a surface-brightness conversion or integrated photometry in v0.1.

8.7 Canonical lineage reconstructibility

Let $\mathcal{L}_o$ denote one canonical calibration-lineage record per coherent reduction package. For every valid occurrence, it must be possible to resolve

$$\begin{aligned} \mathcal{L}_o \longrightarrow \{ & x_j^{xs}, b_j, K^{\text{acq}}, R^{\text{src}}, E^{\text{assoc}}, R^{\text{child}}, G^{\text{src}}, T^{\text{child}}, \\ & D^{\text{del}}, F^{\text{sel}}, \mathcal{O}_a, \tau_{225}, X, C_a, Q, \Pi_o, \Omega_o, Q_o^{\text{owner}}, \\ & \mathcal{V}, \theta, P_B, L \} \longrightarrow M_j \longrightarrow y_j. \end{aligned} \quad (28)$$

The arrows mean resolvable scientific lineage, not a required file layout. Compact product links may point to the canonical record and immutable APT; they need not duplicate dense tables in every product.

9 Normative scientific requirements

The words “must”, “must not”, “required”, and “invalid” are normative. These requirements define scientific conformance without prescribing a file, class, function, language, or storage layout.

9.1 Scope, state, and input typing

SCI-CAL-REQ-001: measured-channel scope.

The v0.1 operation must accept only the ordinary **xs** measured stream. A physically distinct measured stream must be rejected as unsupported and must not inherit the **xs** factor, unit, response, or uncertainty label.

SCI-CAL-REQ-002: target-unit scope.

The only calibrated target is top-of-atmosphere point-source-peak mJy beam^{-1} at $X_{\text{ref}} = 0$, referenced to 272, 214, and 150 GHz for a1100, a1400, and a2000 respectively. Its source-spectrum convention must remain explicit. SCI-CAL must not apply a second source color correction; target color correction and any cross-array spectral comparison belong to the downstream scientist. An unsupported unit or photometric meaning must yield no calibrated SCI-CAL output, not a relabeled numeric array.

SCI-CAL-REQ-003: complete signal declaration.

Each admitted signal must declare its acquisition identity, pre-CAL unit, sample and detector shape, time/support, and upstream eligibility. Ordinary **xs** must mean the dimensionless fitted KID resonance-frequency change $\Delta f/f_{\text{res}}$, with increasing value corresponding to increasing absorbed optical power. SCI-CAL must perform no additive baseline operation or further normalization. The fitted resonance must be consistent with the active probe tone; an invalid Tune invalidates calibration, while excursion beyond about half a resonance width must be exposed as a high-noise quality condition.

SCI-CAL-REQ-004: explicit semantic states.

Missing, disabled, automatic, unavailable, invalid, engineering-only, and science-qualification-eligible states must be represented explicitly. None may be encoded solely as a zero, unity, negative, non-finite, stale, or undocumented sentinel.

SCI-CAL-REQ-005: one-way resolution.

Requested, effective, observation-resolved, and realized calibration states must remain distinct and flow one way. Realized or compatibility state must not mutate the accepted request, and a new observation must replace its resolved state in full. The owner-decision snapshot is immutable input to resolution; candidate or realized state must not alter its decided or explicitly unavailable dispositions.

9.2 Occurrence-scoped identity and association

SCI-CAL-REQ-006: acquisition key.

Every selected detector occurrence must have an observation-local key containing observation/Tune context, network or interface, and the network-local channel/tone slot. A global column is only a locator.

SCI-CAL-REQ-007: binding evidence.

A sample column may be bound to an acquisition occurrence and selected APT row only by an explicit keyed relation or a verified ordered-row contract that records scope, order definition, cardinality, and proof. Row position without that evidence is invalid.

SCI-CAL-REQ-008: unique coverage.

On the declared selected support, binding and target-to-source association must be single-valued and complete. Missing, duplicate, ambiguous, or contradictory keys must invalidate exactly the affected detector or the predeclared required support; no arbitrary choice is permitted.

SCI-CAL-REQ-009: immutable source and selected-child APT identity.

The source APT and selected observation-specific child APT must each be identified as one immutable artifact by resolvable artifact identity and digest. Each used row must carry its artifact-local key and occurrence context. Parent and child remain distinct even for an identity transformation or numerically unchanged value.

SCI-CAL-REQ-010: association evidence.

Each admitted target-to-source edge must name both occurrence-scoped endpoints, matcher method/version, disposition, and available quality evidence. Only an admitted unique match may calibrate; abstained, ambiguous, rejected, or missing edges must not. The record must identify TolProj as the association/child-transform authority and must not assign source-factor meaning to the matcher or consumer.

SCI-CAL-REQ-011: layered identity.

Acquisition, measured-APT, association, and design identities, as well as occurrence, semantic-content, and byte-transport identities, must remain distinct. A design match is not required for ordinary measured Beammap calibration, and a design-ID change alone must not change a valid result.

SCI-CAL-REQ-012: array/network consistency and lifetime.

Network-to-array membership and target/source network context must be checked as declared constraints. Reordering is harmless only when the binding proof remains valid. No identity or row state may leak from another observation.

9.3 Factor authority, causality, and reconstructibility**SCI-CAL-REQ-013: absolute factor.**

The sole absolute detector factor must be the selected child-APT row's finite, nonzero **flxscale**, with direction mJy beam^{-1} per unit **xs**, detector recipient, validity, reference plane, uncertainty status, and selected-artifact lineage declared. Its frozen SCI-BEAM source-calibration record must establish

$$F^{\text{source}} = F_{\text{TOA,nom}}/A_{\text{TOA-equivalent}}.$$

The record must resolve producer identity, source APT/row, calibrator model and epoch, source-atmosphere treatment, finite-source-convolved effective-core fit and support, nominal-beam normalization, acceptance, passband/spectrum, factor direction, and uncertainty/covariance status. A presently absent source **flxscale** uncertainty must remain explicitly unavailable. Reciprocal, sign, or missing generating meaning must never be inferred.

SCI-CAL-REQ-014: pointing-correction disposition.

By default TolProj must select the temporally closest scientifically accepted source APT; no universal maximum age or same-Tune, loading, focus, or detector-state requirement may be invented. At the reducing scientist's direction, TolProj may create a child APT with one per-array photometric rescale based on a suitable recent independent ALMA, SMA, planet, or well-calibrated asteroid reference. The same factor applies to every admitted detector in that array. The child lineage must record source, authorities and epochs, inferred and reference fluxes, spectral convention, factor, parent and child occurrences, and uncertainty when available. An embodied factor must not be applied again; its presently undefined uncertainty mechanism must be marked unavailable, not zero.

SCI-CAL-REQ-015: once-only application.

Absolute **flxscale**, target atmosphere correction, and the explicit target-unit identity must each have realized application count one on every valid occurrence. Parent factors and already embodied events must have runtime count zero.

SCI-CAL-REQ-016: canonical multiplier.

The signal must obey Equations (12) and (13) on exactly $\mathcal{J}_{\text{valid}}$. Scalar multiplication order may change without changing values, but the causal lineage and factor-instance identities must be preserved.

SCI-CAL-REQ-017: responsivity boundary.

responsivity must not be presented or applied as an independent absolute mJy beam^{-1} factor. Any use must have a distinct relative role, unit, normalization/recipient, support, and lineage, and must not duplicate information already embodied in **flxscale**.

SCI-CAL-REQ-018: sensitivity boundary.

Beammap **sens** must not multiply the calibrated signal or masquerade as atmosphere correction or total uncertainty. If used for an approximate weight, that role, units, estimator identity, approximation, support, and exclusions must be explicit.

SCI-CAL-REQ-019: compatibility factor boundary.

An **fcf** or other compatibility aggregate may be admitted only after exact decomposition into the canonical role, unit, direction, constituent instance identities, and exclusions. An opaque aggregate has no calibration authority.

SCI-CAL-REQ-020: factor and package lineage.

Every realized factor must record definition, role, unit, direction, recipient/support, reference plane, application stage, validity, uncertainty status, and parent lineage. One canonical record per coherent reduction package must distinguish the Beammap/source-APT producer, TolProj association/child transformer, TolTECA delivery, and SCI-CAL consumer; bind the Q01–Q09 snapshot, photometric-convention state, and pipeline boundary; and reconstruct Equation (28). Product-level links may resolve it without duplicating the complete APT.

9.4 Atmosphere operator, support, and quality

SCI-CAL-REQ-021: opacity and airmass meaning.

τ_{225} must be the finite nonnegative zenith optical depth measured by the LMT WVR and recorded in the observation's `tel*.nc` stream with time support. X must be the full eligible-sample airmass with model identity/domain. Equation (4) and $X_{\text{ref}} = 0$ must be used; a zenith value alone is not a line-of-sight correction.

SCI-CAL-REQ-022: temporal support.

The approximately five-minute WVR readings must be interpolated in time to the detector sample grid. Every calibrated sample must be covered by an exact opacity state or a declared interpolation between valid bracketing states. Method, version, endpoints, and validity interval must be recorded. Unbracketed interpolation, endpoint extrapolation, and inherited prior- observation opacity are prohibited.

SCI-CAL-REQ-023: complete operator identity.

A numeric atmosphere evaluation requires the exact content-bound operator `am12_fixed_djf25_pieewise_linear_los_tau_v1`, its contract and node digests, array assignment, ordered coordinate/ordinate nodes with units, correction orientation, interpolation rules, numeric support, fixed `LMT_DJF_25.amc` provenance, and frozen TolTECA-v1 passband identity. The operator must fail closed outside its declared support.

SCI-CAL-REQ-024: structural evaluation.

The retained operator must obey Equations (5) through (8): energy-weighted broadband integration by composite trapezoid on the exact supplied nodes, analytic zero anchor, PCHIP interpolation in elevation at each opacity anchor, piecewise-linear interpolation of line-of-sight optical depth across opacity anchors, exact node recovery, and continuity at seams.

SCI-CAL-REQ-025: physical-direction invariants.

On declared positive-opacity support, transmission and correction must be finite and positive, transmission must be strictly below unity and decreasing, and correction must be strictly above unity and increasing. A finite unity plateau, sign change, non-finite value, or direction reversal is invalid.

SCI-CAL-REQ-026: domain intersection.

A calibrated sample must satisfy Equation (10). No opacity, elevation/airmass, line-of-sight coordinate, or time request may be silently clamped, extrapolated, or replaced by a legacy selector/fallback.

SCI-CAL-REQ-027: science-qualification eligibility policy.

CAL must assign one deterministic class to the entire observation. The nominal science-observation guidance is $\tau_{225} \leq 0.15$, with a tolerance of 0.025 for momentary excursions so that they do not invalidate the whole observation. CAL must record the classifier identity, opacity summary, excursions, and decision. The class is operational guidance, not an atmosphere-fidelity, observational-performance, `science-qualified`, or `calibrated-science` claim.

SCI-CAL-REQ-028

Engineering-observation policy. Observations may receive an `engineering-only` class through the same CAL classifier, with experience-based operation commonly extending to $\tau_{225} \simeq 0.25$. Supported samples are calibrated with the same numerical operator and must not be relabeled as science-quality evidence. There is no alternate engineering calibration law.

SCI-CAL-REQ-029: invalid and outside-supported opacity.

Negative or non-finite opacity is invalid input. Opacity above 0.25, absent opacity, invalid airmass/elevation meaning, or a request beyond any operator node/time/model support is outside supported calibration and must yield no calibrated output for the affected sample. Brief unsupported samples do not by themselves invalidate the otherwise supported observation, and the 0.025 classification tolerance never authorizes numerical extrapolation.

SCI-CAL-REQ-030: coherent quality state.

Every observation must receive exactly one opacity quality class; CAL must not split one observation into science and engineering pieces or switch class sample by sample. The 0.15 and 0.25 values are flexible experience-based limits, not physical or error discontinuities, and their policy may change only with scientific-owner approval.

SCI-CAL-REQ-031: passband limits.

Where the operator depends on a passband, it must bind the exact selected TolTECA-v1 passband-set identity, use the supplied throughput as an energy-weighted response, and integrate on its exact nodes without spectral extrapolation. The authorized reference-spectrum surfaces are $\alpha \in \{-1, 0, 2, 4\}$, default $\alpha = 0$, with no interpolation or extrapolation in α . One modeled array-average curve is used; detector/network variation

and passband uncertainty remain unavailable.

9.5 Conditional and nuisance uncertainty

SCI-CAL-REQ-032: downstream conditional covariance boundary.

Measurement-noise and statistical-covariance estimation are downstream of CAL. Equation (17) states the consistency rule a downstream companion must obey when it is expressed on the same ordered support; CAL must not claim to have estimated that covariance.

SCI-CAL-REQ-033: scalar variance and weight boundary.

CAL must not create a statistical variance or weight. A downstream diagonal variance or inverse variance on the same support must be consistent with Equation (18), with units $(\text{mJy beam}^{-1})^2$ or $(\text{mJy beam}^{-1})^{-2}$. Weight is not support, validity, hits, response, confidence, or total uncertainty.

SCI-CAL-REQ-034: unavailable conditional uncertainty.

A CAL product must state that conditional measurement uncertainty is downstream and unavailable from CAL. It must not synthesize zero variance, infinite weight, or a value derived from `sens`.

SCI-CAL-REQ-035: nuisance completeness ledger.

The lineage must carry an entry for detector `flxscale`, common absolute calibrator scale, any TolProj pointing-derived correction, WVR/atmosphere model, Beammap `sens` when used for approximate weights, and beam/template response. The present source-APT `flxscale`, TolProj rescale, WVR/atmosphere, and cross-array covariance mechanisms are not yet numerically available. Each entry must be `quantified`, `not_applicable` with justification, or `unavailable`, and name detector, array, observation, cohort, or global correlation scope when known.

SCI-CAL-REQ-036: correlation preservation.

Nuisance covariance must preserve declared common and cross terms. Shared calibrator, array, opacity, pointing, passband, or beam terms must not be treated as independent per sample or counted down solely with sample or detector number. Systematic scale terms are common within an array. WVR state is observation-common across arrays but produces array-dependent response; numerical cross-array covariance is presently unavailable.

SCI-CAL-REQ-037: total-uncertainty claim.

Total calibrated uncertainty or significance is available only when every required nuisance category and material cross-covariance is quantified or explicitly shown not applicable. Otherwise it must be labeled unavailable, even when conditional weight exists.

SCI-CAL-REQ-038: propagation regime.

Linear nuisance propagation may be claimed only inside one differentiable operator interval with sufficiently small approximately symmetric uncertainties and justified dependence assumptions. Seam-spanning, large, asymmetric, selection, association, or same-data fitted-state uncertainty requires nonlinear or joint propagation; missing cross terms may not be set to zero without justification.

SCI-CAL-REQ-039: shared realized operator.

Signal, conditional covariance/weights, admitted synthetic injections, conditional noise realizations, and genuine Jacobian companions must use the identical realized multiplier, valid support, and factor lineage. A different measured stream must not be substituted as a companion. SCI-CAL acts before PTC; PTC owns DC and correlated atmospheric/common-mode removal.

9.6 Response basis, validity, and claims

SCI-CAL-REQ-040: originating response basis.

Every calibrated result must resolve its producer-owned Beammap source, source APT, selected child APT, and per- detector beam/template occurrence. Available elliptical major/minor axes, position angle, units, coordinate frame, epoch/support, fit identity, and uncertainty status must be retained.

SCI-CAL-REQ-041: realized response distinction.

Originating Beammap response and realized mapmaker/kernel/filter response must remain separate provenance objects. Circularization must be explicitly labeled with its rule and cannot silently replace available elliptical information.

SCI-CAL-REQ-042: point-source-peak condition.

A downstream result may claim literal point-source-peak mJy beam^{-1} only when its realized response satisfies

Equation (26) with $\rho_L = 1$ for the declared beam/template and support, or when an explicit response renormalization establishes and records that condition.

SCI-CAL-REQ-043: response-changing operations.

A kernel, filter, or map operation that changes peak response must propagate the response basis and factor. Without a separately established renormalization it must withhold the literal peak-calibration claim; SCI-CAL itself must not certify the downstream response.

SCI-CAL-REQ-044: excluded photometric meanings.

V0.1 must not emit or authorize MJy/sr, Jy/pixel, Rayleigh-Jeans or thermodynamic temperature, extended-source calibration, integrated photometry, or a resolved-source recovery claim.

SCI-CAL-REQ-045: validity tuple and causes.

Every coherent result must publish the tuple in Equation (1) and machine-distinguishable reason codes for missing identity, unsupported stream/unit, invalid or out-of-domain atmosphere, intentionally disabled calibration, unavailable conditional or nuisance uncertainty, incomplete response basis, and any required output failure. The associated owner-decision snapshot must make every Q01–Q09 disposition and any explicitly unavailable numerical product machine-distinguishable without converting it into a numeric sentinel.

SCI-CAL-REQ-046: truthful no-output states.

Disabled/uncalibrated, invalid, engineering-only, outside-supported calibration, and uncertainty-unavailable states must remain distinguishable. A failure in a required factor or identity must not emit a calibrated number; a missing non-signal uncertainty may leave the signal valid only with the corresponding claim explicitly unavailable. A supported engineering sample may carry calibrated signal under the same operator but must not acquire a science-quality claim.

SCI-CAL-REQ-047: canonical lineage record.

One resolvable canonical record per coherent reduction package must include raw observation identity, source and selected-child APTs and digests, acquisition and row occurrences, association edge, approved child transformation, delivery identity, factor definition and generating record, atmosphere/operator/passband identities and support, reference plane, target unit, photometric-convention state, pipeline boundary/order state, Q01–Q09 snapshot, state tuple, uncertainty ledger, and response basis.

SCI-CAL-REQ-048: compact product links.

Every calibrated product must contain links and occurrence keys sufficient to resolve its canonical lineage, exact source APT, selected child APT, and governing owner-decision snapshot. It need not and should not duplicate the entire APT or dense per-detector lineage table in every product.

SCI-CAL-REQ-049: separated claim status.

Structural correctness, atmosphere representation fidelity, relative repeatability, and absolute flux performance must be reported as separate claims with separate evidence and support. A pass in one layer must not be promoted to another. In particular, **science-qualification-eligible** must not be promoted to an achieved **science-qualified** or **calibrated-science** claim. Unless the owner later accepts the measured evidence, either achieved claim requires separately reported atmosphere-representation fidelity and observational performance over the same declared identity and support in addition to structural correctness. End-to-end closure and independent calibration transfer must remain distinct claims.

SCI-CAL-REQ-050: falsifiability before qualification.

Validation must follow the realizable Beammap workflow: create and `toltec_beammap`-accept source APTs, enter them into `apt_library`, associate them through TolProj, and reduce the matched observation by the ordinary science path. Applying an APT to its generating Beammap tests end-to-end closure, not independent absolute accuracy. Applying it to the associated pointing observation tests calibration transfer only when an adequate independent ALMA, SMA, planet, or well-calibrated asteroid flux is available. Available tests must be reported separately by array, identifying observations, source and child APTs, TolProj associations, references, atmosphere, recovered/input ratios or residuals, repeatability, absolute recovery, and condition dependence. No arbitrary minimum sample size or combinatorial matrix is required. The provisional one-percent, five-percent, and five-to-ten-percent values are reporting benchmarks, not automatic pass/fail ceilings. Final scientific acceptance is an explicit owner decision based on honestly and accurately reported evidence.

10 Limiting cases, pathologies, and falsifiable predictions

The following predictions are consequences of the common authority. They are pre-validation statements, not reports of executed evidence. “No output” means no value carrying a calibrated SCI-CAL unit label; a diagnostic or explicitly uncalibrated passthrough may exist only as a different product state.

Table 5: Required limiting and pathological outcomes.

ID	Challenge	Falsifiable prediction
SCI-CAL-EDGE-001	Zero opacity	For every supported array and finite admitted airmass, $\ell = 0$, $T = C = 1$, and atmosphere correction changes neither signal nor conditional uncertainty. With every other structural condition satisfied, this may establish only science-qualification-eligible ; it does not establish an achieved science-qualified or calibrated-science claim without the separately accepted evidence in SCI-CAL-REQ-049 .
SCI-CAL-EDGE-002	Increasing opacity	At fixed admitted $X > 0$, increasing positive τ_{225} strictly decreases T and increases C while remaining finite. Any finite positive-opacity unity plateau is a failure.
SCI-CAL-EDGE-003	Zenith and airmass	At zenith $X = 1$, $\ell = \tau_{225}$. At fixed positive opacity, a larger admitted airmass gives larger ℓ and larger correction. At zero opacity the result is independent of airmass.
SCI-CAL-EDGE-004	Exact atmosphere node	Evaluation at $\ell_{a,k}$ returns the declared authoritative $-\ln T^{\text{eff}}$ ordinate exactly, followed by the declared exponential orientation. Interpolation of transmission or correction instead is detectable at nontrivial intervals.
SCI-CAL-EDGE-005	Operator seam	Left and right values agree at every node. Slopes may differ; a derivative or linear covariance claim exactly at, or spanning, a seam is unavailable and requires nonlinear propagation.
SCI-CAL-EDGE-006	Invalid operator domain	A value outside numeric node support, outside the upstream airmass model, or outside declared time support yields no calibrated output. Clamping, extrapolation, and a legacy fallback are detectable failures.
SCI-CAL-EDGE-007	Bracketed opacity series	A sample between two valid source states is admissible only if the named source interpolation rule authorizes that bracket and its gap. Its lineage resolves both endpoints, method, and validity interval.
SCI-CAL-EDGE-008	Unbracketed opacity series	A sample before the first state, after the last state, across a disallowed gap, or covered by only one endpoint is outside temporal support. It yields no calibrated output rather than an endpoint hold or extrapolation.
SCI-CAL-EDGE-009	Engineering opacity	Any unsplit segment containing $0.15 < \tau_{225} \leq 0.25$ may receive one observation-level engineering-only class. Every numerically supported sample uses the same calibration operator, while the class conveys no science-quality claim.
SCI-CAL-EDGE-010	Outside or malformed opacity	Negative or non-finite opacity is invalid. Opacity above 0.25 or absent opacity is outside supported calibration for the affected sample. Neither case is silently clamped or relabeled, but a momentary unsupported sample does not automatically invalidate other supported samples in the observation.
SCI-CAL-EDGE-011	Unity factor	Replacing one required factor value by numeric unity leaves values unchanged only when unity is the intended value; the factor role and application count remain present and auditable.
SCI-CAL-EDGE-012	Omitted, duplicated, or inverted factor	For an isolated nonzero factor f , output ratios are exactly those in Equation (15). Signal and uncertainty discrepancies identify the challenged role.

ID	Challenge	Falsifiable prediction
SCI-CAL-EDGE-013	Corrected-APT double application	With $F^{\text{sel}} = PF^{\text{parent}}$, applying P again changes the result by P and violates application count. Using the parent without the embodied event changes it by P^{-1} . An unapproved, unidentified, or out-of-domain child transformation yields no calibrated output even when its numeric value is plausible.
SCI-CAL-EDGE-014	Keyed row permutation	Arbitrarily permuting selected APT rows while preserving row occurrence keys and the explicit keyed relation leaves every output, validity state, and nuisance recipient invariant.
SCI-CAL-EDGE-015	Ordered-row permutation	Permuting rows under an ordered-row binding without supplying a new proof invalidates the binding. A newly proven permutation may be admitted and then gives the keyed-equivalent result.
SCI-CAL-EDGE-016	Network or column reorder	Reordering network containers or sample columns while preserving acquisition keys and bindings leaves results invariant after restoring occurrence order. Inferring identity from the new position produces a detectable mismatch or failure.
SCI-CAL-EDGE-017	Missing or duplicate key	A missing acquisition/APT key, duplicate keyed candidate, inconsistent array/network constraint, or multiple sample columns for one selected occurrence invalidates the affected declared scope. No first-row or nearest-row choice is allowed.
SCI-CAL-EDGE-018	Association disposition	A unique admitted match may calibrate. Abstained, ambiguous, rejected, and missing associations produce no calibrated value for the endpoint; aggregate matcher performance cannot convert one into a match. The matcher cannot become the owner of source-factor meaning, and an association without an exact selected-child artifact is incomplete.
SCI-CAL-EDGE-019	Irrelevant design identity	Adding, removing, or changing a design-ID match while measured occurrence endpoints, association, selected APT values, and all other state remain fixed leaves the SCI-CAL result unchanged.
SCI-CAL-EDGE-020	Scalar conditional uncertainty	Multiplying signal by M multiplies variance by M^2 and divides inverse variance by M^2 , including for a signed F^{sel} . Support and validity do not change.
SCI-CAL-EDGE-021	Vector conditional covariance	For a dense positive-definite input covariance, the output equals AC_xA^T , including all off-diagonal signs and magnitudes. Replacing it with a diagonal is not the same claim.
SCI-CAL-EDGE-022	Unavailable uncertainty	Removing a conditional uncertainty input leaves it unavailable, not zero. Removing any required nuisance entry marks total calibrated uncertainty unavailable while a separately valid signal may remain.
SCI-CAL-EDGE-023	Common systematic and sample count	A shared fractional scale term gives Equation (23). Repeating or averaging identically affected samples does not reduce that fractional term as $N^{-1/2}$.
SCI-CAL-EDGE-024	Nonlinear nuisance case	Large/asymmetric factor uncertainty, uncertainty crossing an operator seam, or uncertain discrete association causes first-order propagation to be withheld. Posterior sampling or a bounded sensitivity calculation need not agree with a local linearization.
SCI-CAL-EDGE-025	Unresolved point source	Under a unit-peak originating beam and a realized response with $\rho_L = 1$, an unresolved source of flux density S mJy has ideal peak S mJy beam $^{-1}$. This is a response prediction, not an integrated-flux or target-color-corrected spectral claim. The result retains the declared per-array reference-spectrum convention until the downstream scientist applies a target color correction.

ID	Challenge	Falsifiable prediction
SCI-CAL-EDGE-026	Response-changing kernel	If a downstream kernel changes ρ_L to r , the unrenormalized ideal peak is rS mJy beam ⁻¹ . A literal peak claim requires recorded renormalization by r^{-1} and an updated response basis.
SCI-CAL-EDGE-027	Circularized beam	Replacing available elliptical axes by a circular beam without an explicit rule is nonconformant. With a declared circularization, the result is labeled approximate and does not erase the originating ellipse.
SCI-CAL-EDGE-028	Unsupported stream or target	A non- xs stream, or a request for surface brightness, per-pixel flux, temperature, extended-source calibration, or integrated photometry, yields an unsupported no-output state without a calibrated v0.1 label. A stream that is not the dimensionless fitted KID resonance-frequency change cannot acquire calibrated meaning merely from the xs token.
SCI-CAL-EDGE-029	Zero signal and additive convention	With an explicit ordinary $x^{xs} = 0$, every finite multiplier gives $y = 0$: CAL creates no additive offset. Any nonzero additive CAL term is a detectable violation; DC and correlated-mode removal belong to downstream PTC.
SCI-CAL-EDGE-030	Observation-order replay	Two observations with distinct source and child APTs, associations/transformations, delivery records, atmosphere, photometric/order states, owner-decision snapshots, and response identities give identical per-observation results in either processing order. No state or factor from one appears in the other's canonical lineage.

10.1 Proposed evidence, separated by claim

No scientific validation is executed by this authorship task. Evidence must follow the realizable TolTEC calibration workflow; it must not invent an arbitrary combinatorial matrix or minimum sample size.

- V1. Analytic structural fixtures.** Use small scalar and dense-vector cases to evaluate Equations (3), (7), (12), (17), and (18); include every edge in Table 5. Exact identities and metadata are compared exactly; numeric tolerances are preregistered for the arithmetic precision and never absorb a semantic mismatch.
- V2. Metamorphic identity and lineage fixtures.** Permute keyed rows, sample columns, and network containers; vary irrelevant design identity; challenge missing/duplicate keys and every association disposition; process observations in both orders; and inject omitted, duplicated, and inverted factor instances. Expected invariants and ratios are fixed by Equations (14) and (15).
- V3. Atmosphere structural fixtures.** Verify the content-bound operator's commit and digests, exact nodes, analytic zero anchor, positivity, strict direction, seam continuity, and support rejection without using observational data. This establishes structural correctness only.
- V4. Atmosphere representation report.** Compare the finite retained operator with its named higher-fidelity generating calculation on a dense, reproducible grid over the exact declared support, using the exact selected passband identity and a fully declared weighting convention. Report maximum and distributional error separately by array, opacity, and airmass. The prior one-percent value is a reporting benchmark, not an automatic pass/fail ceiling. Unknown passband uncertainty remains a limitation.
- V5. Beammap closure test.** Reduce calibrated Beammaps to source APTs, have `toltec_beammap` establish that they are scientifically acceptable, enter them into `apt_library`, use TolProj's ordinary matching machinery to associate each APT back to its generating Beammap, and reduce that Beammap through the ordinary science path. Report recovered/input flux ratios or residuals separately for each array. This establishes end-to-end numerical closure of APT-TolProj-CAL-downstream processing, not independent absolute calibration accuracy.
- V6. Associated-pointing transfer test.** Use TolProj to apply an accepted Beammap-derived APT to that Beammap's associated pointing observation and reduce it through the same ordinary science path. When the pointing source has an adequate independent ALMA, SMA, planet, or well-calibrated asteroid flux, this tests calibration transfer. Across the observations actually available, record observation and source/child

APT identities, TolProj association, reference and recovered flux, array, atmosphere, flux ratio or residual, repeatability, absolute recovery, and observed condition dependence. The five-percent and five-to-ten-percent values are reporting benchmarks only. Final scientific acceptance is an owner decision based on the honestly reported evidence.

11 Evidence package for a future audit or clean implementation

A review package can remain implementation neutral while exposing enough evidence to falsify the contract. The recommended evidence inventory is:

- a content-bound Q01–Q09 owner-decision snapshot plus claim-by-claim fixtures demonstrating that each explicitly unavailable product and final owner-acceptance state cannot be fabricated by candidate metadata;
- immutable input and resolved-state records for at least two observations with deliberately different detector subsets, selected APTs, associations, atmosphere states, and response bases;
- source-calibration, TolProj association/child-transform, TolTECA delivery, and SCI-CAL consumption records, including identity-transform, approved-transform, out-of-domain-transform, and no-reinterpretation cases;
- the exact operator and passband content identities, node inventory, orientation, support, and opacity interpolation records;
- broadband-compatibility and pipeline-boundary records covering the exact reference frequencies, authorized spectra, CAL-before-PTC order, and downstream target color correction;
- a machine-resolvable factor ledger showing role and application count, including one corrected-APT ancestry challenge;
- downstream scalar and dense-vector covariance-consistency fixtures, plus systematic-ledger cases at detector, array, observation, cohort, and global scopes, including current unavailable states;
- identity metamorphic cases covering keyed permutations, proven and broken ordered relations, network reorder, missing/duplicate keys, every association disposition, and irrelevant design-ID changes;
- response fixtures for an unresolved source before and after a known response-changing kernel; and
- output inventories, canonical lineage, compact links, validity tuples, reason codes, and claim-layer status for every case, including every truthful no-output state.
- Beammap closure records that trace source APT creation and `toltec_beammap` acceptance through `apt_library`, TolProj matching, ordinary science reduction, and recovered/input flux by array; and
- associated-pointing transfer records, when adequate independent ALMA, SMA, planet, or well-calibrated asteroid fluxes exist, identifying the observations, source and child APTs, associations, references, atmosphere, array, flux ratios/residuals, repeatability, absolute recovery, and observed condition dependence.

For each case, the evidence states expected equations and edge IDs before execution, numeric precision and tolerances, exact identity expectations, and whether a mismatch invalidates a detector, segment, or entire required output. A skipped required case remains unavailable. Structural fixtures do not stand in for the separately proposed atmosphere representation and observational studies in Section 10.1.

12 Conformance disposition

A future reviewer can use four non-overlapping dispositions:

Table 6: Disposition meanings.

Disposition	Meaning
Structurally conformant	All applicable numbered requirements and predeclared structural predictions pass on the exact reviewed candidate and support; this says nothing stronger about atmosphere fidelity or performance.
Nonconformant	One or more applicable equations, requirements, identity or support invariants, uncertainty transforms, lineage conditions, or truthful state outputs fail.
Unavailable	Required authority or evidence is absent. This is the current claim-specific disposition for the current source-APT, TolProj-rescale, WVR/systematic, cross-array-covariance, total-uncertainty, and unaccepted performance products; it is not a pass or a failure of an untested implementation.
Not applicable	A requirement’s premise is demonstrably absent and the canonical record gives the scientific justification, for example a pointing- correction premise shown to be absent and marked <code>not_applicable</code> .

Atmosphere representation fidelity and observational performance receive separate dispositions and evidence. Their provisional numerical targets never convert “unavailable” into “pass”.

13 Engineering conclusion

The engineering conformance surface is finite and falsifiable: dimensionless `xs` input, layered identity, unique association, producer-owned source calibration, approved child transformation, immutable delivery, selected-child factor authority, once-only composition, supported atmosphere evaluation, broadband and CAL-before-PTC order, downstream uncertainty consistency, truthful quality/validity, response-basis preservation, and package-level reconstructibility. No current implementation mapping is part of this view.

The exact numerical atmosphere record is now bound, as are Q01–Q09’s scientific semantics. Conformance therefore requires using that record rather than choosing nodes, orientation, support, or a fallback inside an implementation. Remaining unavailable uncertainty products must be reported as unavailable, not used to suppress a valid signal and not represented as zero. Structural conformance, atmosphere representation, Beammap closure, and independent pointing-source transfer remain separate evidence layers. The provisional one-percent, five-percent, and five-to-ten-percent values are reporting benchmarks only. Final **science-qualified** or **calibrated-science** acceptance is an owner decision based on the honestly and accurately reported evidence.